

Hard University-Level Practice MCQs

Propositional Logic, Boolean Algebra, Canonical Forms, Karnaugh Maps and Sequences

Difficulty: approximately 9/10 Questions: 50 Suggested time: 120 minutes

Instructions: Select the single best answer for each question. The answer key is provided at the end.

Propositional Logic

1. Let p , q , and r be propositions. Which expression is logically equivalent to $\neg(p \rightarrow (q \wedge r))$?

- A. $\neg p \vee (q \wedge r)$
- B. $p \wedge (\neg q \vee \neg r)$
- C. $p \wedge \neg q \wedge \neg r$
- D. $\neg p \wedge (q \vee r)$

2. Which proposition is a tautology?

- A. $((p \rightarrow q) \wedge p) \rightarrow q$
- B. $(p \rightarrow q) \wedge (q \rightarrow p)$
- C. $(p \vee q) \rightarrow (p \wedge q)$
- D. $(p \rightarrow q) \rightarrow (p \wedge q)$

3. The statement “A student may enter the laboratory only if the student has an ID card and wears safety shoes” is best represented by E = enter, I = has ID, S = wears safety shoes as:

- A. $(I \wedge S) \rightarrow E$
- B. $E \rightarrow (I \wedge S)$
- C. $E \leftrightarrow (I \wedge S)$
- D. $(I \vee S) \rightarrow E$

4. Which expression is logically equivalent to $(p \leftrightarrow q)$?

- A. $(p \vee q) \wedge (\neg p \vee \neg q)$
- B. $(p \wedge q) \vee (\neg p \wedge \neg q)$
- C. $(p \wedge \neg q) \vee (\neg p \wedge q)$
- D. $(p \rightarrow q) \wedge (\neg p \rightarrow q)$

5. For how many assignments of p , q , r is $(p \wedge q) \rightarrow r$ false?

- A. 1
- B. 2
- C. 3
- D. 4

6. Which pair is NOT logically equivalent?

- A. $p \rightarrow q$ and $\neg q \rightarrow \neg p$
- B. $\neg(p \vee q)$ and $\neg p \wedge \neg q$
- C. $p \leftrightarrow q$ and $(p \rightarrow q) \vee (q \rightarrow p)$
- D. $\neg(p \rightarrow q)$ and $p \wedge \neg q$

7. The compound proposition $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is:

- A. A contradiction
- B. A contingency
- C. A tautology
- D. Equivalent to $p \wedge q \wedge r$

8. If p is false, q is true, and r is false, what is the truth value of $(p \vee \neg q) \leftrightarrow (r \rightarrow q)$?

- A. True
- B. False
- C. Cannot be determined
- D. Depends only on p

9. Which expression is the negation of “If the server is online, then either the database is available or the backup is active”?

- A. Server offline and database unavailable and backup inactive
- B. Server online and database unavailable and backup inactive
- C. Server online and either database unavailable or backup inactive
- D. Server offline or database unavailable or backup inactive

10. How many rows are true in the truth table of $(p \oplus q) \vee r$, where $\oplus$ denotes exclusive OR?

- A. 3
- B. 4
- C. 5
- D. 6

Boolean Algebra

11. Simplify $F = A + A'B + AB'$.

- A. $A + B$
- B. $A + B'$
- C. $A' + B$
- D. $A \oplus B$

12. Simplify $F = (A + B)(A + B')$.

- A. A
- B. B
- C. $A + BB'$
- D. AB

13. Using De Morgan's law, the complement of $F = A(B + C')$ is:

- A. $A' + B'C$
- B. $A'B'C$
- C. $A' + B' + C$
- D. $A' + BC'$

14. Simplify $F = AB + A'C + BC$.

- A. $AB + A'C$
- B. $AB + BC$
- C. $A'C + BC$
- D. $A + C$

15. Simplify $F = (A + B + C)(A + B + C')$.

- A. $A + B$
- B. $A + BC$
- C. $AB + C$
- D. $A + B + CC'$

16. Which expression implements XOR using only AND, OR and NOT?

- A. $AB + A'B'$
- B. $A'B + AB'$
- C. $(A + B)'$
- D. $A' + B'$

17. If $F = (A \text{ NAND } B) \text{ NAND } (A \text{ NAND } B)$, then F equals:

- A. $A + B$
- B. AB
- C. $(AB)'$
- D. $A \oplus B$

18. A NOR gate with both inputs connected to A produces:

- A. 0
- B. 1
- C. A
- D. A'

19. Simplify $F = A'B'C + A'BC + AB'C + ABC$.

- A. A
- B. B
- C. C
- D. $A + B + C$

20. Simplify $F = (A + B')(A' + B')(A + B)$.

- A. AB
- B. $A'B'$
- C. AB'
- D. $A + B'$

21. Which identity is valid for all Boolean variables?

- A. $A + AB = AB$
- B. $A(A + B) = A$
- C. $A + A'B = AB$
- D. $(A + B)(A' + B) = A + B$

22. The dual of $A + 0 = A$ is:

- A. $A \cdot 0 = A$
- B. $A \cdot 1 = A$
- C. $A + 1 = 1$
- D. $A' + 0 = A'$

23. Simplify $F = A(B + C) + A(B + C)'$.

- A. A
- B. $B + C$
- C. $A(B + C)$
- D. 1

24. Simplify $F = (A' + B)(A + B)$.

- A. $A + B$
- B. B
- C. AB
- D. $A'B$

25. For $F = A \oplus B \oplus C$, $F = 1$ when:

- A. Exactly one input is 1 only
- B. An even number of inputs are 1
- C. An odd number of inputs are 1
- D. All three inputs are equal

Canonical Forms

26. For variables ordered A, B, C, the minterm $A'BC$ corresponds to:

- A. m_2
- B. m_3
- C. m_5
- D. m_6

27. For variables ordered A, B, C, the maxterm M_5 is:

- A. $A + B' + C$
- B. $A' + B' + C$
- C. $A' + B + C'$
- D. $A + B + C'$

28. If $F(A,B,C) = \sum m(1,2,5,7)$, which canonical POS is correct?

- A. $\prod M(1,2,5,7)$
- B. $\prod M(0,3,4,6)$
- C. $\prod M(0,1,2,3)$
- D. $\prod M(3,4,6,7)$

29. The canonical SOP of $F = A + BC$, using variables A, B, C, is:

- A. $\sum m(3,4,5,6,7)$
- B. $\sum m(1,2,3,7)$
- C. $\sum m(0,3,4,7)$
- D. $\sum m(3,5,6,7)$

30. Which minterm is 1 only for $A=1, B=0, C=1, D=0$?

- A. $AB'CD'$
- B. $A'BC'D$
- C. $AB'C'D$
- D. $ABCD'$

31. If $F(A,B,C) = \prod M(0,2,6)$, then F' is:

- A. $\sum m(0,2,6)$
- B. $\prod M(1,3,4,5,7)$
- C. $\sum m(1,3,4,5,7)$
- D. $\prod M(0,2,6)$

32. A Boolean function of four variables has 1s in exactly 10 truth-table rows. Its canonical POS contains:

- A. 6 maxterms
- B. 10 maxterms
- C. 16 maxterms
- D. 4 maxterms

33. Which expression is a canonical SOP term for four variables A, B, C, D?

- A. $A'BC$
- B. $(A + B' + C + D)$
- C. $AB'C'D$
- D. $AB + CD$

34. $F(A,B,C)=A'B+AC$ has which minterm expansion?

- A. $\Sigma m(2,3,5,7)$
- B. $\Sigma m(1,2,3,7)$
- C. $\Sigma m(2,3,4,5)$
- D. $\Sigma m(1,3,5,7)$

Karnaugh Maps

35. The minimum SOP for $F(A,B,C)=\Sigma m(1,3,5,7)$ is:

- A. $A + B + C$
- B. C
- C. C'
- D. $AB + C$

36. The minimum SOP for $F(A,B,C)=\Sigma m(0,2,4,6)$ is:

- A. C'
- B. C
- C. B'
- D. A'

37. For $F(A,B,C,D)=\Sigma m(0,2,8,10)$, the minimum SOP is:

- A. $B'D'$
- B. $A'C'$
- C. $C'D'$
- D. $B'C'$

38. For $F(A,B,C,D)=\Sigma m(0,1,2,3,8,9,10,11)$, the minimum SOP is:

- A. A'
- B. B'
- C. C'
- D. D'

39. Which set of four minterms forms one valid 4-cell K-map group?

- A. $\{0,1,2,4\}$
- B. $\{0,1,4,5\}$
- C. $\{0,3,12,15\}$
- D. $\{1,2,5,6\}$

40. $F(A,B,C,D)=\Sigma m(1,3,5,7,9,11,13,15)$ simplifies to:

- A. A
- B. B
- C. C
- D. D

41. $F(A,B,C,D)=\Sigma m(0,1,4,5,8,9,12,13)$ simplifies to:

- A. C'
- B. D'
- C. B'
- D. A'

42. In a minimal SOP K-map solution, a variable is eliminated from a group when:

- A. It is 0 in every cell
- B. It is 1 in every cell
- C. It changes value within the group
- D. It appears complemented

43. For $F(A,B,C,D)=\Sigma m(0,1,2,3,4,5,6,7)$, the minimum expression is:

- A. A
- B. A'
- C. B'
- D. C'

Sequences

44. An arithmetic sequence has $a_5=18$ and $a_{12}=46$. What is a_1 ?

- A. 0
- B. 2
- C. 4
- D. 6

45. Which explicit formula generates the arithmetic sequence 11, 7, 3, -1, ...?

- A. $a_n = 15 - 4n$
- B. $a_n = 11 - 4n$
- C. $a_n = 7 - 4n$
- D. $a_n = 4n + 7$

46. Evaluate Σ from $k=1$ to 6 of $(2k - 1)$.

- A. 21
- B. 30
- C. 36
- D. 42

47. Evaluate Π from $k=2$ to 5 of $(k - 1)$.

- A. 12
- B. 24
- C. 60
- D. 120

48. If $a_1=3$ and $a_n=a_{n-1}+2n$ for $n \geq 2$, what is a_5 ?

- A. 23
- B. 27
- C. 31
- D. 33

49. Given $a_n=3n-2$, evaluate Σ from $n=1$ to 10 of a_n .

- A. 135
- B. 145
- C. 155
- D. 165

50. An arithmetic sequence has common difference 5 and the sum of its first 20 terms is 1070. What is its first term?

- A. 4
- B. 5
- C. 6
- D. 7

Answer Key

1-B 2-A 3-B 4-B 5-A 6-C 7-C 8-B 9-B 10-D
11-A 12-A 13-A 14-A 15-A 16-B 17-B 18-D 19-C 20-C
21-B 22-B 23-A 24-B 25-C 26-B 27-C 28-B 29-A 30-A
31-A 32-A 33-C 34-A 35-B 36-A 37-A 38-B 39-B 40-D
41-A 42-C 43-B 44-B 45-A 46-C 47-B 48-C 49-B 50-C

Coverage follows the uploaded lecture material: propositional equivalence and truth tables; Boolean laws and universal gates; minterms, maxterms, SOP/POS; K-map grouping; and sequence, summation, product, factorial and arithmetic-sequence concepts.